

Computer Exercise-01: We start from Step-4

Open TABMATE, create a new TABLO file [file/new/TAB file], and write the following code with the above structure. When writing this TAB file, make sure you understand what it is. Especially, in equation part, please type it in reference to equation in Step-2 slide.

Tips: in the middle of your typing, press F4 once in a while, to check whether you already type correctly (for syntak error).

!————— START OF TABLO FILE —————!

FILE

MDATA # data file #;

SET

IND (i1,i2);

VARIABLE

(all,i,IND) x(i) # output of domestic industry #;

(all,i,IND) l(i) # labor in domestic industry #;

(all,i,IND) k(i) # capital in industry #;

(all,i,IND) p(i) # price of commodity produced by industry #;

(all,i,IND) q(i) # household demand for commodity #;

w # nominal wage #;

r # rent of capital in industry #;

y # household total income #;

ls # household supply of labor #;

ks # household supply of capital #;

(all,i,IND) c(i) # unit-cost or effective price of primary factor #;

COEFFICIENT

(all,i,IND) VX(i) # value of production of industry #;

(all,i,IND) VL(i) # labor cost of industry #;

(all,i,IND) VK(i) # industry capital #;

(all,i,IND) SL(i) # Share of labor costl #;

VLS # value of labor supply #;

VKS # value of capital supply #;

VY # value of total income #;

(all,i,IND) SIGMA(i);

READ

VL from file MDATA header "VL";

VK from file MDATA header "VK";

UPDATE

(all,i,IND) VL(i) = w*l(i);

(all,i,IND) VK(i) = r*k(i);

FORMULA

(all,i,IND) SIGMA(i) = 1;

```

(all,i,IND) VX(i) = VL(i) + VK(i);
(all,i,IND) SL(i) = VL(i)/VX(i);
VLS = sum{i,IND,VL(i)};
VKS = sum{i,IND,VK(i)};
VY = VLS + VKS;

```

EQUATION

```

e_l # demand for labor #
(all,i,IND) l(i) = x(i) - SIGMA(i)*[w - c(i)];
e_k # demand for capital #
(all,i,IND) k(i) = x(i) - SIGMA(i)*[r - c(i)];
e_c # price of primary factor composite #
(all,i,IND) c(i) = SL(i)*w + (1 - SL(i))*r;
e_pf # zero profit in production #
(all,i,IND) VX(i)*[p(i) + x(i)] = VL(i)*[w + l(i)] + VK(i)*[r + k(i)];
e_ls # labor market clearing #
VLS*ls = sum{i,IND,VL(i)*l(i)};
e_mc # market clearing for commodities #
(all,i,IND) x(i) = q(i);
e_y # household factor income #
VY*y = VLS*(w + ls) + VKS*(r + ks);
e_q # demand for commodity #
(all,i,IND) q(i) = y - p(i);

```

!————— END OF TABLO FILE —————!

Note that we remove one of the market clearing equation (capital market) by Walras' Law.

Save your first tablo file as modelA.tab, and press the Tablo Code icon (implementing the tablo file¹).

Definition 1 *Implementing the Tablo file is creating some files needed to run simulation. This is done by clicking the icon "Tablo Implement" or by pressing F11. After creating a new or modifying model, you have to implement the tablo file to prepare the model for later simulation, otherwise your modification is ignored.*

Step 5. Creating the database file

Database file record the 'base' coefficient, here the coefficients are WL_i and RK_i . In GEMPAK the database file has the extension .har, and created and read by a GEMPAK program called VIEWHAR.

0.1 Computer Exercise

1. Open VIEWHAR program.
2. Create New File (File Menu)
3. Create New Header (Edit Menu)

¹See Definition.

- For WL_i , the header name is "VL", number of dimension is 1, dimension is 2.
 - Do the same for RK_i .
4. Move the value of WL_i and RK_i from the EXCEL file (io.xls), by copy and Paste.
Note: Given the structure of the I-O file, industry is in column, and factor (capital and labor) is in row, you have to transpose it first in EXCEL before you can paste it to VIEWHAR.
 5. Save the file as modelA.har.

Step 6. Creating simulation file

Simulation file is a text file with extension .CMF where you put the closure² and shock statement, and some other ancilliary information.

Definition 2 Closure. *Closure is a stement of which variables are endogenous, and which variables are exogenous that make sure that the number of endogenous variables is equal to the number of statement and also will tell us what variables that can be shocked.*

You will create a simulation file. Open TABMATE program, create new CMF file [File/New/CMF File], and type the following.

!———— START OF SIMULATION FILE —————!

```
Auxiliary file = modelA;
File MDATA = modelA.har;
Updated File MDATA = modelA.upd;
Solution file = modelA ;
!- CLOSURE —!
exogenous ks ls p("i1");
Rest endogenous;
Verbal Description = Model A;
Method = Johansen;
!- SHOCK STATEMENT —!
shock p("i1") = 10;
```

!———— END OF SIMULATION FILE —————!

Save the file as simA.cmf.

In the simulation file, here, we explicitly list the exogenous variables i.e. k^S, l^S , and w . By exogenizing w , we set price of labor as the numeraire, and we want to check the model, whether or not, it pass the homogeneity test³, by change the numeraire by 10%.

Definition 3 Homogeneity Test. *Homogeneity test is changing the numeraire by certain percentage, when the model is correctly specified, with a correct database, then all nominal variables will change by the same proportion, while all real (quantity) variables remain unchanged.*

²See Definition.

³See Definition.

Definition 4 Real Homogeneity Test. Testing the property of constant returns to scale technology. If all input (factor supply) are increased by certain percentage, then, all real variables will change by the same proportion, leaving all prices unchanged.

Step 7. Simulation

Now, the simulation is carried out, with the following steps,

1. Open the WINGEM program.
2. Change the working directory to the directory/folder where the Tablo file (*.TAB), database file (*.HAR), and simulation file (*.CMF) are saved.⁴
3. Run GEMSIM Solve (Simulation Menu)
4. Select the command file i.e. simA1.cmf
5. Click 'run'.
6. Click 'Go to ViewSOL', where you can see the result of the simulation.
7. Check whether or not, the model pass homogeneity test.

Examples of simulations and Exercises

After conducting the last step, you all have the practical experience of the CGE modelling from A to Z with the simple model. Now, we will shock k^S by 10%, and analyze the result.

Increase in capital endowment (k^S)

It is always better, to think about what we expect from the result of the simulation. Figure 2.1.a illustrate the effect of increase in capital endowment.

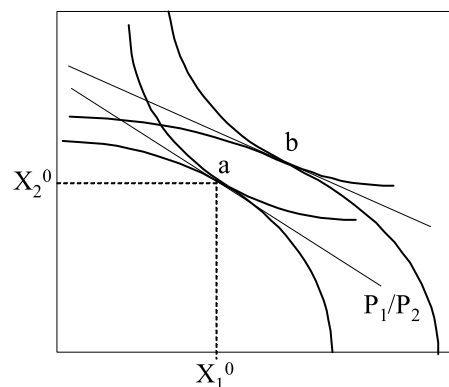


Figure 2.1.a

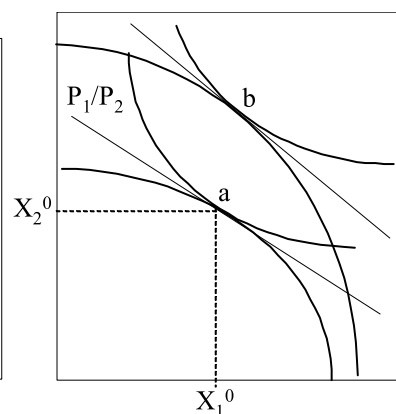


Figure 2.1.b

Since the database says that industry 1's (manufacturing) capital share in its total cost is higher than industry 2's (non-manufacturing) we could say that industry 1 is more capital intensive⁵. We can then expect the following

⁴Tips. Create the name of the folder as simple as possible, and in the location that is easy to reach such as d:\modelA\.

⁵**Small Query.** Since in microeconomics factor intensity is usually define as input ratio in quantity i.e. K/L , how cost share can imply factor intensity in this model?

- Industry 1's output will increase more proportionately than industry 2's output (including their demand at equilibrium).
- Price ratio (P_1/P_2) will drop, the slope of price ratio will be lower, or industry 1's price will drop more proportionately than price of industry 2.
- Increase in supply of kapital relative to labor will reduce rent relative to wage, or factor price ratio (W/R) will rise.

After doing the simulation by adding statement of "shock ks = 10;" to the CMF file, complete the table below.

Result of increasing k^S by 10%

Variable		Industry	
		i1	i2
k	k_i		
l	l_i		
p	p_i		
q	q_i		
c	c_i		
x	x_i		
ks	k^S		
ls	l^S		
r	r		
w	w		
y	y		

Answer the following question,

1. Open your general equilibrium exercise with MathCAD and check whether it gives you the same results.
2. Look at the change in output, and compare between the two industry, is it consistent with what you expect (the theory)?
3. Look at the commodity price result, is it also consistent?
4. What happen to relative factor prices (W/R)?
5. Bearing in mind your answer to question 3, what do you think will happen to capital labor ratio (K/L) in both industry? [Hint: this is substitution effect, the change in capital labor ratio with respect to change in relative factor price]. Can you confirm from the result, that substitution effect do take place?
6. What happen to ouput of industry 2, what happen to its labor use? Can you explain what's going on, given industry 2 is labor intensive industry?
7. What happen to welfare/utility?
8. What happen to income? Is it contradictory to your answer to question 6?
9. Can you calculate inflation? If yes, by howmuch? Can you relate this to question 7? [Hint: Calculate real income]
10. Can you think of how to calculate real GDP?

Tablo Exercise

1. Modify your model file (ModelA.TAB) to add variable and equation that define inflation (cpi)!

2. Modify your model file (ModelA.TAB) to add variable and equation that define real GDP!

Real homogeneity test.

Shock all real variables, here k^S and l^S by the same percentage change. Since we have constant returns to scale technology, all the real variables must change by the same amount, while all nominal variables are unchanged.